Notation.
Let $X: U \to \mathbb{R}^3$ be a patch. Define

$$g_{ij} \overset{\text{def}}{=} \langle X_i, X_j \rangle$$

The first-fundamental form $I$ on $M$ is

$$I: T_pM \times T_pM \to \mathbb{R}: (A, B) \mapsto \sum_{1 \le i,j \le 2} g_{ij} \cdot A^i \cdot B^j$$

And, $g \overset{\text{def}}{=} \det(g_{ij})$ and $g^{kl} = (k,l)$ entry of $(g_{ij})^{-1}$.

The normal of $X$ at $X(s)$ is ($s = (u,v)$)

$$n(s) = \frac{X_1(s) \times X_2(s)}{\| X_1(s) \times X_2(s) \|}.$$

And, the second-derivatives at $X(s)$ is

$$X_{ij}(s) = \frac{\partial^2 X}{\partial u^j \partial u^i}(s)$$

And, the instrinsic normal of a unit speed curve in the image of $X$ is

$$S = n \times T.$$

The Normal and Geodesic Curvature of a unit speed curve $\gamma$ in $X$

$$k_n = \langle \gamma''(s), n(\gamma'(s), \gamma^2(s)) \rangle \text{ and } k_g = \langle \gamma''(s), S(s) \rangle$$

The coefficients of the second fundamental form of $X$ is

$$L_{ij} = \langle X_{ij}, n \rangle \quad (1 \le i, j \le 2)$$

The Second-fundamental form $II$ on $M$ is the bilinear form

$$II: T_pM \times T_pM \to \mathbb{R}: (X, Y) \mapsto \sum_{1 \le i,j \le 2} L_{ij} \cdot X^i \cdot Y^j$$

with associated Matrix $L$. ($L^l_k = \sum L_{ik} \cdot g^{il}$)

The Christoffel symbols

$$\Gamma^k_{ij} \overset{\text{def}}{=} \sum_{l=1}^{2} \langle X_{ij}, X_l \rangle g^{lk}.$$

Debris

So that given two tangent vectors $A = \sum a^i X_i$ and $B = \sum b^j X_j$ satisfies

$$\langle A, B \rangle = \sum_{1 \le i,j \le 2} a^i b^j \cdot \langle X_i, X_j \rangle = \begin{pmatrix} b^1 & b^2 \end{pmatrix} \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \cdot \begin{pmatrix} a^1 \\ a^2 \end{pmatrix}$$

Forms of the Monge patch.

Let $X: U \to \mathbb{R}^3 : (u^1, u^2) \mapsto (u^1, u^2, f(u^1, u^2))$ where $f$ is a differentiable.

Then, $\begin{cases} X_1 = (1, 0, f_1) \\ X_2 = (0, 1, f_2) \\ n = (-f_1, -f_2, 1)/\sqrt{1 + (f_1)^2 + (f_2)^2} \end{cases}$ ← It forms the 1st form.

and

$$\begin{cases} X_{11} = (0, 0, f_{11}) \\ X_{12} = (0, 0, f_{12}) = (0, 0, f_{21}) = X_{21} \\ X_{22} = (0, 0, f_{22}) \end{cases}$$

The associated Matrix of the 1st-form is

$$\begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} = \begin{bmatrix} 1 + (f_1)^2 & f_1 f_2 \\ f_1 f_2 & 1 + (f_2)^2 \end{bmatrix} \quad (\text{so that } I(A, B) = {}^t B \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \cdot A)$$

The associated Matrix of the 2nd-form is

$$\begin{bmatrix} L_{11} & L_{12} \\ L_{21} & L_{22} \end{bmatrix} = \frac{1}{\sqrt{1 + (f_1)^2 + (f_2)^2}} \cdot \begin{bmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{bmatrix}$$ ← hess(f).

And the Christoffel symbols is

$$\Gamma_{ij}^{k} = \frac{f_{ij} \cdot f_k}{1 + (f_1)^2 + (f_2)^2}$$

Full calculation of the Christoffel symbol.

$$
\begin{aligned}
\Gamma_{11}^{1} &= \langle X_{11}, X_1\rangle \cdot g^{11} + \langle X_{11}, X_2\rangle \cdot g^{21} \\
&= \langle X_{11}, X_1\rangle \cdot g_{22} \cdot (\det g_{ij})^{-1} + \langle X_{11}, X_2\rangle \cdot (-g_{21}) \cdot (\det g_{ij})^{-1} \\
&= \left(f_{11} \cdot f_1 \cdot \left[1 + (f_2)^2\right] + f_{11} \cdot f_2 \cdot (-f_1 f_2)\right) / \left(1 + (f_1)^2 + (f_2)^2 + (f_1 f_2)^2 - (f_1 f_2)^2\right) \\
&= f_{11} \cdot f_1 / \left(1 + (f_1)^2 + (f_2)^2\right) \\
\Gamma_{11}^{2} &= \langle X_{11}, X_1\rangle \cdot g^{12} + \langle X_{11}, X_2\rangle \cdot g^{22} \\
&= \langle X_{11}, X_1\rangle \cdot (-g_{12}) \cdot (\det g_{ij})^{-1} + \langle X_{11}, X_2\rangle \cdot g_{11} \cdot (\det g_{ij})^{-1} \\
&= \left(f_{11} \cdot f_1 \cdot (-f_1 f_2) + (f_{11} \cdot f_2) \cdot \left[1 + (f_1)^2\right]\right) / \left(1 + (f_1)^2 + (f_2)^2\right) \\
&= f_{11} \cdot f_2 / \left(1 + (f_1)^2 + (f_2)^2\right) \\
\Gamma_{12}^{1} &= \langle X_{12}, X_1\rangle \cdot g^{11} + \langle X_{12}, X_2\rangle \cdot g^{21} \\
&= \langle X_{12}, X_1\rangle \cdot g_{22} \cdot (\det g_{ij})^{-1} + \langle X_{12}, X_2\rangle \cdot (-g_{21}) \cdot (\det g_{ij})^{-1} \\
&= \left(f_{12} \cdot f_1 \cdot \left[1 + (f_2)^2\right] + f_{12} \cdot f_2 \cdot (-f_1 \cdot f_2)\right) / \left(1 + (f_1)^2 + (f_2)^2\right) \\
&= f_{12} \cdot f_1 / \left(1 + (f_1)^2 + (f_2)^2\right) \\
\Gamma_{12}^{2} &= \langle X_{12}, X_1\rangle \cdot g^{12} + \langle X_{12}, X_2\rangle \cdot g^{22} \\
&= \left(f_{12} \cdot f_1 \cdot (-f_1 f_2) + f_{12} \cdot f_2 \cdot \left[1 + (f_1)^2\right]\right) / \left(1 + (f_1)^2 + (f_2)^2\right) \\
&= \left(f_{12} \cdot \left[f_2 + (f_1)^2 \cdot f_2 - (f_1)^2 \cdot f_2\right]\right) / \left(1 + (f_1)^2 + (f_2)^2\right) \\
&= f_{12} \cdot f_2 / \left(1 + (f_1)^2 + (f_2)^2\right). \\
\Gamma_{22}^{1} &= \langle X_{22}, X_1\rangle g^{11} + \langle X_{22}, X_2\rangle \cdot g^{21} \\
&= \left(f_{22} f_1 \cdot \left[1 + (f_2)^2\right] + f_{22} \cdot f_2 \cdot (-f_1 f_2)\right) / \left(1 + (f_1)^2 + (f_2)^2\right) \\
&= f_{22} \cdot f_1 / \left(1 + (f_1)^2 + (f_2)^2\right) \\
\Gamma_{22}^{2} &= f_{22} \cdot f_2 / \left(1 + (f_1)^2 + (f_2)^2\right)
\end{aligned}
$$

Shape Operator.

Def. The map

$$L: T_pM \to \mathbb{R}^3: X \to -X_n$$

(where $n: D(\subset M) \to S^2: q \mapsto n(q)$ and $Xf$ is a notation of directional derivative).
is called the shape operator. In actually, $L \in \mathrm{End}(T_pM)$.

Lemma. Let $M$ be a surface and $L: T_pM \to \mathbb{R}^3: X \to -X_n$ be the shape operator.

$$L(a\cdot X_1 + b\cdot X_2) = \begin{bmatrix} L_{11}\cdot g^{11} + L_{21}\cdot g^{21} & L_{12}\cdot g^{11} + L_{22}\cdot g^{21} \\ L_{11}\cdot g^{12} + L_{21}\cdot g^{22} & L_{12}\cdot g^{12} + L_{22}\cdot g^{22} \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}$$

$$= \begin{bmatrix} L_1^1 & L_2^1 \\ L_1^2 & L_2^2 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} \quad \text{where } L_k^l = L_{1k}\cdot g^{1l} + L_{2k}\cdot g^{2l}$$